

Piotr Koper

Senior Data Engineer · Platform & Infrastructure

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PROFESSIONAL SUMMARY

Senior Data Engineer with 4+ years at Jaguar Land Rover, specialising in data platform infrastructure, CI/CD pipeline engineering, and cloud infrastructure on GCP. Progressed from pipeline development and data products into technical leadership — currently leading working groups across GitLab CI/CD, Terraform, and Docker image management. Background in logistics and operations management across major UK brands provides strong domain knowledge and stakeholder communication skills developed at scale. Open to Data Engineering, Platform Engineering, DevOps, and Analytics Engineering roles where infrastructure depth and engineering rigour are valued.

TECHNICAL SKILLS

Programming Languages: Python · SQL · Bash · HCL · YAML

Cloud Platforms & Services: GCP — BigQuery · GCS · Cloud Run · Cloud Functions · Cloud SQL · Artifact Registry · AWS · Snowflake

Infrastructure & IaC: Terraform · Docker · Kaniko · Podman · Kubernetes · Helm

CI/CD & DevOps: GitLab CI/CD · GitHub Actions · GitLab Components · Pipeline Templating · Container Scanning

Data Orchestration & Transformation: Dagster · Apache Airflow (Cloud Composer / Astronomer) · dbt · Dataform

Pipeline Patterns: ETL / ELT · Pipeline Architecture · Data Modelling · Data Contracts · APIs

Security & Identity: Workload Identity Federation · IAM · Container Scanning · Secret Management

Developer Tooling: Git · Linux · Pre-commit · uv · Grafana

PROFESSIONAL EXPERIENCE

Senior Data Engineer · [Jaguar Land Rover](#) · 2022 — Present

Part of the core Data Engineering team, focused on infrastructure, developer tooling, and engineering standards. Progressed from pipeline development and data products into platform engineering, taking on technical leadership across multiple working groups.

Data Pipelines & Products

- Designed, developed and delivered data pipelines and data products across the organisation, applying Data Contract standards to enforce schema validation and detect data anomalies as part of the pipeline quality framework

CI/CD Working Group — Lead

- Led refactoring and standardisation of GitLab CI/CD pipeline templates across the data engineering organisation, reducing pipeline duplication and establishing a reusable component library adopted across all teams
- Designed authentication template supporting both Workload Identity Federation and legacy Service Account Key methods — ensuring full backward compatibility across all existing projects with zero disruption to live pipelines

Workload Identity Federation — Owner & Lead

- Sole owner of end-to-end WIF rollout between GitLab and GCP — designed and provisioned full Terraform infrastructure (Pool, Provider, bindings) and redesigned the CI/CD authentication layer
- Eliminated service account key sprawl across the data engineering organisation, improving security posture and reducing credential management overhead
- Produced comprehensive documentation covering architecture, implementation, and migration guide — adopted as the organisation-wide standard

Docker Images Working Group — Lead

- Led design, maintenance, and build of custom Docker images stored in GCP Artifact Registry, used across data engineering pipelines and services
- Implemented container scanning and image governance standards, reducing security vulnerabilities in pipeline execution environments

Terraform Working Group — Lead

- Lead the Terraform working group overseeing module standards, provisioning patterns, and IaC practices across the data platform, improving consistency and reducing infrastructure provisioning time

Notifications System

- Core developer of GCP-based notifications system triggering messages to correct channels based on subscription behaviour and API endpoint calls during pipeline execution
- Fully refactored Terraform module for Cloud Run and PostgreSQL infrastructure; authored extensive documentation and operational runbooks adopted by the wider team

GitLab Instance Migration

- Part of team designing and provisioning Kubernetes cluster, namespace, and Helm module for GitLab runners during a full instance migration — all infrastructure managed via Terraform

Python Task Orchestrator Evaluation — Lead

- Led structured technical evaluation of enterprise Python task orchestrators across two platforms, building end-to-end data pipelines including data validation, CI/CD integration, event-triggered execution, and cross-platform authentication
- Assessment covered local and cloud deployment, developer experience, private package management, and integrations with GCP, dbt, Dataform, and Snowflake — formal report adopted as recommendation to the Enterprise Technology Board

CPAT Lead & Planner · [Jaguar Land Rover](#) · 2018 — 2022

Cross-functional programme role managing Capacity, Parts Availability and Timing assessment process across JLR vehicle programmes.

- Built redesigned CPAT Input Matrix addressing existing error states, improving data input control and reducing submission errors across programme teams
- Developed automated WIP file creation process saving approximately 12 hours per file — equivalent to 1.5 working days per programme cycle
- Developed eCAR data scrape tool integrated into WIP file workflow, eliminating a previously manual weekly data extraction process

Planning Manager & Logistics Operations · [Halfords](#) · [Aston Martin Lagonda \(DHL\)](#) · [DHL](#) · 2007 — 2018

Over a decade of progressive operations management across major UK distribution and manufacturing environments — Halfords (1.4M units/week), Aston Martin Lagonda, and DHL.

- Consistently built tooling and automation to replace manual processes — Logistics Performance Tool tracking volumes, hours and costs against budget; automated planning tools; redesigned operational workflows
- Managed teams of up to 130 people; selected for competitive DHL Head Start management programme — one of 28 accepted nationally from hundreds of UK and Ireland applicants

PERSONAL PROJECTS

Two personal data engineering projects built on a shared infrastructure pattern: Dagster-based pipelines, PostgreSQL for run state, Docker Compose orchestration, and automated deployment to Oracle Cloud VM via GitHub Actions — build, push to container registry, SSH deploy, and restart with zero manual steps.

Solis Battery Manager · github.com/p-y-t-h-e-c/solis_energy_manager · 2026

- Dagster scheduled pipeline ingesting solar inverter data from the Solis Cloud API (HMAC-signed auth), validating the response, and delivering daily battery charge, PV generation, and export summaries via Pingram API email notification
- Multi-platform Docker images (amd64/arm64) built and published to GitHub Container Registry on every push to main
- Stack: Dagster · Python · Docker · PostgreSQL · GitHub Actions · uv · Pingram

Octopus Energy Dashboard · [Personal Project](#) · 2023

- Dagster pipeline ingesting variable tariff rates and consumption data from the Octopus Energy API, storing in PostgreSQL, and surfacing via Grafana dashboard with day-on-day and year-over-year cost comparisons
- Stack: Dagster · Python · PostgreSQL · Grafana · GitHub Actions · Docker

EDUCATION

MSc Management and Production · [Technical University of Łódź, Poland](#) · 2001 — 2007

Majoring in Systems and Information Technologies

MSc Information and Communication Technology for Engineers · [Coventry University, UK](#) · 2005 — 2006

Analysis and Design of Engineering Information Systems · Project Management · Software Design for Engineers

CERTIFICATIONS & LANGUAGES

GCP Professional Data Engineer · Google Cloud · 2021 (expired — experience remains current)

Polish — native · English — fluent